JAMES CHRISTOPHER HEWITT

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github.com/j-hewitt

PROFESSIONAL SUMMARY

Machine Learning Engineer with 4 years of experience building and deploying production ML systems across fintech and e-commerce sectors. Specialise in Python-based data pipelines, model optimisation, and taking projects from experimentation through to live implementations. I've worked with cross-functional teams to deliver solutions that measurably improved business metrics, and I'm particularly interested in exploring efficient model architectures and interpretability in complex systems.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM

Data Engineering: Pandas, NumPy, Apache Spark, SQL Server, PostgreSQL

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform

MLOps: Docker, Git, Jenkins CI/CD, Model monitoring

Specialisations: Predictive modelling, NLP, Time series forecasting, Feature engineering

WORK EXPERIENCE

Senior Machine Learning Engineer

DataFlow Systems Ltd

www.dataflow.co

Bristol, United Kingdom

June 2022 - Present

Developed and deployed 5 production ML models for credit risk assessment, improving default prediction accuracy from 0.76 to 0.84 AUC across 2 million customer records

Architected automated retraining pipeline using Apache Airflow that reduced manual intervention by 80% and ensured models stayed within performance thresholds

Led feature engineering initiative that identified 47 new derived features, resulting in a 12% improvement in model performance with no additional data collection costs

Mentored 2 junior engineers on ML best practices, code review standards, and helped them deliver their first production models within 3 months

Machine Learning Engineer

TechVenture Solutions

www.techventure.io

London, United Kingdom (Remote from Bristol)

January 2021 - May 2022

Built end-to-end recommendation engine for e-commerce platform using collaborative filtering and neural networks, increasing average order value by 8.3% across 500k active users

Created comprehensive data validation framework using Great Expectations that caught data quality issues before they reached model training, preventing 3 major incidents

Optimised inference latency for real-time predictions from 850ms to 120ms through model distillation and quantisation, enabling deployment to edge devices

Implemented automated monitoring dashboard tracking 15+ model performance metrics, with alerts configured to catch performance degradation within 2 hours

Data Scientist

Insight Analytics Ltd

www.insightanalytics.net

Bristol, United Kingdom

August 2019 - December 2020

Developed time series forecasting model for retail inventory prediction using LSTM networks, reducing overstock situations by 22% and saving approximately GBP180k annually

Conducted A/B testing framework analysis for marketing campaigns, delivering 15+ statistical reports that informed product strategy decisions

Built Python ETL pipeline processing 2TB of daily transaction data, transforming raw logs into structured datasets for analysis and model training

Created interactive dashboards using Tableau and SQL that enabled non-technical stakeholders to explore customer behaviour patterns independently

EDUCATION

BSc Computer Science (2:1 Honours)

University of Bristol

Bristol, United Kingdom

September 2015 - June 2019

ADDITIONAL INFORMATION

Public Repositories: github.com/j-hewitt/time-series-forecasting, github.com/j-hewitt/nlp-sentiment-classifier, github.com/j-hewitt/feature-engineering-toolkit, github.com/j-hewitt/model-monitoring-suite

Certifications: AWS Certified Machine Learning Specialty, Google Cloud Associate Cloud Engineer